

Python: Data Types and Variables

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1 Datatypes

Name	Type	Description	Example
Integers	<code>int</code>	Positive or negative whole numbers with no decimal point.	7, 31, 35
Floating Point	<code>float</code>	Represent real numbers and are written with a decimal point.	2.01, 9.990, 1.0
String	<code>str</code>	A collection of characters enclosed in quotes.	"CR7", "messi", "500"
Lists	<code>list</code>	A collection of arbitrary objects.	["Hello world", 100, 2.33]
Tuples	<code>tuple</code>	Like lists but immutable.	(15, 9.45, "Hello")
Dictionaries	<code>dict</code>	Unordered key-value pairs.	{"Name": "James", "Roll": "1808031"}
Sets	<code>set</code>	Unordered collection of unique objects.	{"a", "Z"}
Booleans	<code>bool</code>	Logical values indicating truth or falsehood.	True, False

Table 1: Common Python Data Types

2 Variables

	1000
	1001
	1002
	1003
	1004
	1005
	1006
	1007
	1008
	1009

Figure 1: Memory Layout

- A variable is a reserved memory location used to store values.
- Variables allow programmers to use meaningful names instead of memory addresses.
- A variable must be declared before it is used.
- Python uses dynamic typing.

Rules for Variable Names

- A variable name must start with a letter (A-Z, a-z) or an underscore (_).
- A variable name cannot start with a digit (0-9).
- A variable name can contain only letters, digits, and underscores (A-Z, a-z, 0-9, and _).
- Variable names are case-sensitive. For example, `age`, `Age`, and `AGE` are three different variables.
- A variable name cannot be a Python keyword (e.g., `if`, `for`, `while`, `class`).

Operators in Programming Documentation

Operators

Special Symbols that perform different operations.

The value that the operator operates on is called the **operand**.

Types of Operators

1. Arithmetic operators
2. Relational Operators
3. Logical operators
4. Bitwise operators
5. Assignment operators
6. Special operators

Arithmetic operators

Name	Operator	Description
Addition	+	Adds values on either side of the operator.
Subtraction	-	Subtracts right hand operand from left hand operand.
Multiplication	*	Multiplies values on either side of the operator.
Division	/	Divides left hand operand by right hand operand
Exponentiation	**	Performs power calculation on operands.
Modulus	%	Divides left hand operand by right hand operand and returns remainder.
Floor division	//	The digits after the decimal point are removed from result.

Name	Operator	Description	Syntax
Greater than	>	True if left operand is greater than the right	x > y
Less than	<	True if left operand is less than the right.	x < y
Equal to	==	True if both operands are equal	x == y
Not Equal to	!=	True if both operands are not equal	x != y
Greater or Equal	>=	True if left operand is greater than or equal to the right.	x >= y
Less or Equal	<=	True if left operand is less than or equal to the right.	x <= y

Name	Operator	Description	Syntax
Logical AND	and	True if both the operands are true	x and y
Logical OR	or	True if either of the operands is true	x or y
Logical NOT	not	True if operand is false	not x

Relational Operators

Logical operators

Bitwise Operators

Assignment operators

Special operators

Membership Operators

Identity Operators

Complete Operator Summary

[[[1]]]

Category	Operators	Description
Category	Operators	Description
		Continued...

Arithmetic +, -, *, /, **, %, // Mathematical calculations
 Relational >, <, ==, !=, >=, <= Comparison operations
 Logical and, or, not Boolean logic operations

Name	Operator	Description	Syntax
Bitwise AND	&	Sets each bit to 1 if both bits are 1	a & b
Bitwise OR		Sets each bit to 1 if at least one bit is 1	a b
Bitwise XOR	^	Sets each bit to 1 if only one bit is 1	a ^ b
Bitwise NOT	~	Inverts all the bits	~a
Left Shift	«	Shifts bits to the left	a « b
Right Shift	»	Shifts bits to the right	a » b

Bitwise &, |, ^, ~, «, » Bit-level operations

Assignment =, +=, -=, *=, /=, %=, //, **, &=, |=, ^=, «=, »= Assign with operations

Membership in, not in Check presence in sequence

Identity is, is not Check object identity

Name	Operator	Description
Assign	=	Assigns values from right side operands to left side operand.
Add AND	+=	It adds right operand to the left operand and assign the result to left
Subtract AND	-=	It subtracts right operand from the left operand and assign the result
Multiply AND	*=	It multiplies right operand with the left operand and assign the result
Divide AND	/=	It divides left operand with the right operand and assign the result to
Modulus AND	%=	It takes modulus using two operands and assign the result to left oper
Floor Divide AND	//=	It performs floor division and assigns the result to left operand.
Exponent AND	**=	It performs exponentiation and assigns the result to left operand.
Bitwise AND AND	&=	It performs bitwise AND and assigns the result to left operand.
Bitwise OR AND	=	It performs bitwise OR and assigns the result to left operand.
Bitwise XOR AND	^=	It performs bitwise XOR and assigns the result to left operand.
Left Shift AND	«=	It performs left shift and assigns the result to left operand.
Right Shift AND	»=	It performs right shift and assigns the result to left operand.

Operator	Description
in	Evaluates to true if it finds a variable in the specified sequence and false otherwise.
not in	Evaluates to true if it does not finds a variable in the specified sequence and false otherwise

Operator	Description	Example
is	Evaluates to true if both variables refer to the same object.	x is y
is not	Evaluates to true if both variables do not refer to the same object.	x is not y